Cross-Domain Synthesis: Ultrametric Geometry as Common Mathematical Structure Across Quantum Error Correction, Spin Glasses, Protein Folding, Cosmology, and Cognition

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Abstract

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Abstract: We present a cross-domain synthesis demonstrating that ultrametric tree geometry—specifically, cophenetic distance on rooted trees—provides the common mathematical structure underlying hierarchical organization across five disparate domains: quantum error correction, spin glasses, protein folding, cosmology, and cognition. The synthesis is organized around five integration points: (1) triadic rigidity as a universal, falsifiable signature of hierarchical organization; (2) resolution-dependence as a unifying lens that reconciles Archimedean and ultrametric descriptions; (3) bounded consilience (levels L1–L5) as an honesty taxonomy for scoping evidential claims; (4) the noun/verb distinction separating ontic invariants (tree topology) from epistemic conventions (height functions, specific numerical values); and (5) the ultrametric inequality as the mathematical engine from which all other properties derive. We map each integration point onto each of the five domains, with explicit consilience labeling: quantum error correction at L1–L2 (theorem + computational validation with zero logical errors at depth 7 across 36,000 trials), spin glasses at L3 (Parisi's replica symmetry breaking solution), protein folding at L4 (energy landscape theory), and cosmology and cognition at L5 (suggestive structural patterns). We address seven objections and specify domain-specific falsification conditions. The contribution is not new results in any single domain but the connective tissue—the demonstration that the same mathematical signature recurs across domains at varying levels of evidential support, and that this recurrence is neither coincidence nor metaphor.

1. Introduction

1.1 The Geometry-Choice Thesis

Every physical theory makes a choice—usually an implicit one—about what kind of space it inhabits. The standard choice is Archimedean: distances add, space is continuous, and the geometry is Euclidean or pseudo-Riemannian. This choice is so deeply embedded that it rarely surfaces as a *choice*. It presents as obvious, as given.

This paper examines what happens when we make a different choice: when we take the fundamental geometry to be not Archimedean but ultrametric—the geometry of rooted trees, where distance is measured not by addition but by the depth of the lowest common ancestor. The ultrametric choice is not new in isolation. It appears in the replica symmetry

breaking solution to spin glasses (Parisi, 1979), in the energy landscape theory of protein folding (Bryngelson & Wolynes, 1987; Frauenfelder et al., 1991), in taxonomic classification in linguistics and biology, and—most recently—in the computational validation of passive quantum error confinement on Bruhat–Tits trees (Quni-Gudzinas, 2026a, 2026b).

What *is* new is the recognition that these are not isolated instances of tree-like organization. They are manifestations of the same mathematical structure—cophenetic distance on a rooted tree with a monotone height function—recurring across domains at different scales, with different physical realizations, and at different levels of evidential support. The thesis of this paper is:

> **The hierarchical aspects of reality have the mathematical structure of a rooted tree with a monotone height function. This structure is identifiable by a specific, falsifiable signature: triadic rigidity.**

We are not claiming that *everything* is hierarchical. We are claiming that *where* hierarchical organization exists, it has a specific mathematical signature—and that signature is testable.

1.2 Triadic Rigidity: The Universal Signature

The core mathematical result that makes this thesis falsifiable is *triadic rigidity*. For any three items in a hierarchically organized domain, when pairwise distances are measured by cophenetic distance (the height of the lowest common ancestor), the two largest distances must be equal:

$$d_1 = d_2 \geq d_3$$

where d_1, d_2, d_3 are the three pairwise cophenetic distances sorted in descending order.

This is a theorem—it follows necessarily from the ultrametric inequality $d(x, z) \leq \max(d(x, y), d(y, z))$, which cophenetic distance satisfies [L1: mathematical theorem; Quni-Gudzinas, 2026c]. The significance of triadic rigidity is that it converts the ultrametric thesis from a philosophical position ("reality might be tree-like") into a testable mathematical signature: if triadic rigidity fails for a domain, ultrametric structure does not apply. If it holds, the domain is hierarchically organized in the specific sense defined by cophenetic distance.

This is what separates the present synthesis from metaphorical invocations of trees. A metaphor cannot be falsified. Triadic rigidity can.

1.3 Bounded Consilience: What We Claim, and at What Level

A cross-domain synthesis of this scope must be honest about the gradient of evidential support. Not all domains contribute equally. We adopt a five-level consilience taxonomy [L1: mathematical definition; Quni-Gudzinas, 2026c]:

Level	Claim Type	What It Means
L1	Mathematical theorem	Proven from axioms; carries the force of logical necessity
L2	Computational validation	Demonstrated through reproducible numerical experiments by us
L3	Published result (others)	Established in the peer-reviewed literature by independent researchers
L4	Structural analogy	The mathematical structure maps cleanly onto domain phenomena; no contradiction with known results
L5	Suggestive pattern	Observed structural resemblance; consistent with the framework but not yet subjected to rigorous test

Every claim in this paper carries an explicit consilience label [L1]–[L5]. No claim is presented at a level higher than its evidence supports. This is not hedging—it is precision. The framework is strongest where evidence is strongest, and honest about where evidence is suggestive.

The evidence gradient across our five domains is steep:

Domain	Strongest Consilience Level	Nature of Evidence
Quantum error correction	L1 + L2	Theorem + 36,000 computational trials
Spin glasses	L3	Parisi's ultrametric RSB solution
Protein folding	L4	Energy landscape theory
Cosmology	L5	Suggestive structural patterns
Cognition / linguistics	L5	Taxonomic hierarchies, semantic dendrograms

The paper does not collapse this gradient. It uses it: the strong domains (quantum error correction, spin glasses) establish that ultrametric geometry is not merely a mathematical curiosity—it is physically realized. The weaker domains (cosmology, cognition) suggest that the pattern may extend further, providing direction for future investigation without overclaiming.

1.4 Structure of This Paper

Section 2 establishes the mathematical foundation: cophenetic distance, the ultrametric inequality, triadic rigidity, resolution-dependence, the noun/verb distinction, and — in §2.5

— a demonstration that rooted trees are native to number theory through the Farey tree and the Langlands program. Section 3 maps the five integration points onto each of the five domains, with explicit consilience labeling. Section 4 addresses seven objections. Section 5 specifies falsification conditions and outlines next steps.

1.5 A Note on Terminology: From Bruhat–Tits to Cophenetic

Readers familiar with the earlier papers in this program may notice a shift in language. The first computational validations were described as taking place on "Bruhat–Tits trees" — a specific family of $(p+1)$-regular trees arising from the Bruhat–Tits building of $\mathrm{PGL}(2, \mathbb{Q}_p)$ [Quni-Gudzinas, 2026a, 2026b]. The present paper uses the language of "cophenetic distance on rooted trees." This shift is not a correction of an error. It is a generalization — and it is worth explaining explicitly.

The Bruhat–Tits tree T_p **is** a rooted tree. Cophenetic distance — $d(x, y) = h(\mathrm{lca}(x, y))$, the height of the lowest common ancestor — can be defined on it, and every mathematical property derived in this paper (the ultrametric inequality, triadic rigidity, nested equivalence relations, disjoint open balls) holds on T_p as it does on any rooted tree. The computational validations that demonstrated zero logical errors at depth $d = 7$ across $36{,}000$ trials [L2; Quni-Gudzinas, 2026a, 2026b] were performed on specific tree architectures — first binary ($p = 2$), then ternary ($p = 3$) — and those results are unaffected by the change in terminology. The numbers do not care what we call the tree.

What changed is the recognition that the essential properties are **not specific to Bruhat–Tits trees**. The ultrametric inequality and triadic rigidity are properties of cophenetic distance on ANY rooted tree with a monotone height function — a fact proven in the Tree Distance Cophenetic paper [L1; Quni-Gudzinas, 2026c]. This recognition emerged from two developments:

1. **The Symmetric Extension paper [L2; Quni-Gudzinas, 2026b] found that the strict Bruhat–Tits construction introduces structural asymmetry:** the root has $p + 1$ children while internal nodes have p children. At $p = 2$, this asymmetry collapsed the error barrier for logical 1 to a constant $B = 2$. The fix — giving ALL nodes exactly p children — **deviates from the strict Bruhat–Tits definition.** The resulting architecture is a regular p-ary tree, not a Bruhat–Tits tree in the strict sense. The terminology had to shift to reflect what was actually being used.

2. **The Tree Distance Cophenetic paper [L1; Quni-Gudzinas, 2026c] proved that the mathematical engine — the ultrametric inequality, triadic rigidity, resolution-dependence, the noun/verb distinction — applies to ALL rooted trees.** What began as an investigation of a specific number-theoretic object (T_p) was recognized as a general framework applicable to any hierarchically organized domain.

The shift in language is therefore not instability. It is maturation — from a specific instance to the general class. Newton called derivatives "fluxions." We do not conclude that

calculus is unreliable because the terminology evolved. The mathematical facts are invariant under changes in what we call them.

Why this matters for the present synthesis: If the framework were restricted to Bruhat–Tits trees — objects defined over p-adic numbers and tied to the Langlands program — it could not reach the other four domains. Spin glasses are not p-adic. Protein folding funnels are not Bruhat–Tits buildings. Cosmology and cognition are not number-theoretic. By recognizing that the framework is cophenetic distance on rooted trees — not Bruhat–Tits trees specifically — we gain the generality needed for cross-domain synthesis without losing the specific results obtained on the trees we actually used.

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2. Mathematical Foundation

2.1 Cophenetic Distance and the Ultrametric Inequality

Consider a rooted tree T with a height function $h(v)$ that assigns a real number to each node v, monotone decreasing from root to leaves: if u is an ancestor of v, then $h(u) \geq h(v)$. The leaves of the tree represent items in the domain. For any two leaves x and y, their *cophenetic distance* is:

$$d(x, y) = h(\mathrm{lca}(x, y))$$

where $\mathrm{lca}(x, y)$ is the lowest common ancestor of x and y—the deepest node that is an ancestor of both.

The physical intuition is straightforward: two items are close when they share a long common history before diverging (their LCA is deep in the tree, near the leaves). They are far when they diverge early (their LCA is shallow, near the root). The distance measures *how far back toward the root you must travel before the two items part company*.

Cophenetic distance satisfies the *ultrametric inequality* [L1]:

$$d(x, z) \leq \max(d(x, y), d(y, z))$$

This is a stronger condition than the standard triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$. The ultrametric inequality says: for any three items, the two largest pairwise distances are equal. This is triadic rigidity.

The proof follows directly from tree structure [L1; Quni-Gudzinas, 2026c, §2.2]. For any three leaves x, y, z, their pairwise LCAs form a nested hierarchy: among the three LCAs $\mathrm{lca}(x, y)$, $\mathrm{lca}(y, z)$, and $\mathrm{lca}(x, z)$, two must be equal and lie at or above the third. The corresponding distances inherit this equality.

2.2 Resolution-Dependence: Two Rulers, Two Pictures

The height function $h(v)$ is a resolution parameter. The root corresponds to the coarsest resolution (all items are indistinguishable—they share the root as their LCA). Moving

downward, successive branching events differentiate items at finer resolutions. The leaves represent the finest resolution at which each item is a distinct individual.

This reveals that the choice between Archimedean and ultrametric geometry is a *resolution choice*, not a truth claim about the fundamental nature of space [Quni-Gudzinas, 2026d]:

Resolution	What You See
Coarse	Continuous manifold (Archimedean)
Fine	Discrete tree structure (ultrametric)

At coarse resolution, distinctions blur and the discrete tree approximates a continuous manifold—much as a fine-grained photograph looks continuous from a distance. At fine resolution, the discrete branching structure becomes visible. The two descriptions are not competitors. They describe the same system at different scales.

This resolution-dependence is not a philosophical nicety. It has concrete consequences in each domain. In quantum error correction, resolution corresponds to tree depth d—the number of branching levels—and the error barrier scales as $B(d) = 2^d$ [L2; Quni-Gudzinas, 2026a]. In spin glasses, resolution corresponds to inverse temperature—cooling reveals successively finer structure in the state space [L3; Parisi, 1979]. In protein folding, resolution corresponds to the depth of the folding funnel—how finely the energy landscape discriminates between conformations [L4; Bryngelson & Wolynes, 1987].

2.3 Noun/Verb: Ontic vs. Epistemic

A common mistake in cross-domain synthesis is to confuse *what is invariant* (the ontic structure) with *how we measure it* (the epistemic convention). The tree framework provides a clean separation [L1; Quni-Gudzinas, 2026c, §5]:

Category	What It Is	Invariant?
Noun (ontic)	Tree topology—the branching order	Yes—independent of measurement convention
Verb (epistemic)	Height function, specific distance values, energy scales	No—depend on gauge choice, measurement convention, units

The branching order is what persists regardless of how we assign heights. If two items share a deep LCA, they are genuinely close in the tree's intrinsic topology—this is true whether we assign heights in units of energy, time, semantic distance, or any other measure. But the *numerical value* of that closeness is gauge-dependent.

This distinction prevents the most common error in the literature: treating a specific numerical overlap value $q_{\alpha\beta}$ in spin glasses, or a specific energy barrier in protein folding,

or a specific error rate in quantum circuits as *the* invariant. These are epistemic—they depend on the measurement convention. The invariant is the tree structure itself.

2.4 The Ultrametric Inequality as Mathematical Engine

All five integration points derive from a single mathematical fact: cophenetic distance satisfies the ultrametric inequality. From this inequality follow:

1. **Triadic rigidity** (§2.1): $d_1 = d_2 \geq d_3$—the falsifiability mechanism.
2. **Nested equivalence relations**: At each height h, items with $d(x, y) \leq h$ form an equivalence class—a cluster at resolution h.
3. **Strong triangle inequality**: All triangles are isosceles with the base no longer than the equal sides.
4. **Disjoint open balls**: Two open balls of the same radius are either disjoint or identical—there is no partial overlap. This is the geometric origin of error confinement in quantum circuits [L1–L2; Quni-Gudzinas, 2026a, 2026d].

These properties are not imposed. They are consequences of the tree structure. If a domain is organized as a rooted tree with cophenetic distance, these properties hold automatically. If they fail, the domain is not tree-structured.

2.5 Precedent — The Farey Tree and Trees in Number Theory

The rooted tree is not an exotic structure imported into mathematics for the purposes of this synthesis. It is fundamental to number theory itself. The **Farey tree** — a close relative of the Stern–Brocot tree — organizes the rational numbers in the interval $[0, 1]$ into a binary tree hierarchy by denominator size. At the root sits $\frac{1}{2}$. Its children are $\frac{1}{3}$ and $\frac{2}{3}$. Each node $\frac{a}{b}$ produces children via the mediant operation with its Farey neighbors. The result is an infinite binary tree containing every rational number in $(0, 1)$ exactly once, with the denominator acting as a natural depth parameter — larger denominators lie deeper in the tree.

The Farey tree is not merely a convenient visualization. It is a **generative structure**: the rational numbers are produced by the tree, not placed on it after the fact. The branching encodes the arithmetic relationships between fractions. The depth of a rational number in the tree is a function of its denominator, and cophenetic distance — the depth of the lowest common ancestor of two fractions — measures the denominator at which they first diverge. The tree IS the arithmetic [L1: the Farey tree is a standard construction in number theory; Hardy & Wright, 1938].

This tree appears concretely in the theory of **modular forms**. The theta function of the Gaussian lattice $\mathbb{Z}[i]$:

$$\theta_{\mathbb{Z}[i]}(q) = \sum_{x,y \in \mathbb{Z}} q^{x^2+y^2}, \quad q = e^{\pi i \tau}$$

is a modular form. When visualized on the unit circle via $q = e^{2\pi i\theta}$, the circle is partitioned into Farey arcs — arcs between consecutive Farey fractions. At each rational point $\theta = h/k$, the modular form exhibits a **spike** — a cusp-like singularity whose height is given by quadratic Gauss sums [L1: standard result in the theory of modular forms; Iwaniec, 1997].

The Farey tree is literally inscribed in the complex plane by the behavior of modular forms. The spikes can be interpreted as harmonic resonances — points where frequencies lock into rational alignment and constructive interference occurs [L5: suggestive interpretation in physical terms; Quni-Gudzinas, 2026f, §5].

The Langlands Bridge. The Langlands program proposes a systematic correspondence — a "dictionary" — between two domains that were historically treated as separate [L3: Langlands, 1967, 1970; established research program]:

Arithmetic Side	Harmonic Side	Bridge
Galois representations (symmetries of algebraic equations)	Automorphic forms (wave-like functions on symmetric spaces)	L-functions
Prime numbers (building blocks of integers)	Spectra / frequencies (fundamental vibrations)	Zeros of L-functions
Rational points (discrete arithmetic objects)	Resonant peaks (constructive interference in frequency space)	Critical values

The Modularity Theorem — the first major victory of the Langlands program, proved by Wiles, Taylor, et al. — established that every elliptic curve (a geometric object defined over the rationals) is "modular" — it corresponds to a modular form (a wave-like object in harmonic analysis) [L3: Wiles, 1995; Breuil et al., 2001]. Two domains that appeared entirely separate turned out to be the same thing viewed from different sides. The L-function is the bridge — an infinite product over primes that encodes both the arithmetic of the curve and the spectrum of the form.

Why this matters for the present synthesis. The Langlands correspondence is itself a cross-domain bridge of exactly the kind this paper argues for. It connects number theory to harmonic analysis through a common mathematical structure. The Farey tree — a rooted tree with cophenetic structure — sits at the intersection: it organizes the rational points where modular forms resonate, and its branching encodes the arithmetic relationships that Galois representations formalize. The tree is not a metaphor imported into mathematics. It is a native mathematical object, and the Langlands program is a proven instance of the cross-domain unity that this synthesis investigates at broader scale.

In a different direction, the same tree geometry may extend to the level of fundamental constants. The fine-structure constant α can be expressed as a degenerate cross-ratio $\alpha = r_e/\bar{\lambda}_C = (0, \infty; r_e, \bar{\lambda}_C)$, where r_e is the classical electron radius and $\bar{\lambda}_C$ is the

reduced Compton wavelength [L5: suggestive; Quni-Gudzinas, 2026g]. This observation is exact — the ratio is an algebraic identity — but the interpretation of fundamental constants as cross-ratios on a tree-structured parameter space remains speculative. The adelic product formula constraint, which would require dimensionless constants to be rational numbers, is not supported by current measurement of $\alpha^{-1} = 137.035999084\ldots$, which shows no periodic decimal pattern. The boundary between the established cross-ratio identity and the speculative rationality constraint is precisely where further investigation is needed — and the triadic rigidity test, applied to correlations among fundamental constants, offers a potential pathway.

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3. Domain Evidence

3.1 Quantum Error Correction [L1–L2]

Strongest evidence in the synthesis.

The Bruhat–Tits tree T_p is a rooted tree where every node has exactly p children. For $p = 3$ (the minimal symmetric architecture), the tree has p^d leaves at depth d. Physical qubits are placed at the leaves, and the tree's ultrametric geometry provides passive error confinement: errors propagate upward through the tree but are geometrically contained within their branch [L2; Quni-Gudzinas, 2026a, 2026b].

Triadic rigidity test: For any three encoded logical states on the tree, cophenetic distance between their error syndromes satisfies $d_1 = d_2 \geq d_3$. This is the geometric signature of tree-structured error confinement.

Computational validation: The Symmetric Extension paper [L2; Quni-Gudzinas, 2026b] reported: - At depth $d = 7$ ($2{,}187$ leaves): zero logical errors across all physical error rates $p_{\text{err}} \leq 0.40$ (Wilson 95% CI upper bound: 0.0019) - $36{,}000$ total trials with identical error rates for both logical states (symmetric) - Constructive barrier proof through $d = 15$: $B(d) = 2^d$ - The Tier 0 paper [L2; Quni-Gudzinas, 2026a] independently validated the binary tree architecture for logical 0

Barrier scaling: The error barrier $B(d) = 2^d$ follows directly from the ultrametric nesting property: to flip a logical state, an error must overcome every branching level simultaneously. The probability of this decreases exponentially with depth [L1–L2].

Noun/verb in QEC: The tree topology (branching order, $p = 3$ structure) is the ontic invariant. Specific error rates at specific physical error probabilities are epistemic—they depend on the measurement convention. The barrier exponent $B(d) = 2^d$ is L1 (mathematical consequence of tree structure); the zero-error result at finite depth is L2 (computational validation).

3.2 Spin Glasses [L3]

Strongest independent corroboration from outside the QWAV program.

Giorgio Parisi's replica symmetry breaking (RSB) solution to the Sherrington–Kirkpatrick spin glass model demonstrated that the equilibrium state space is ultrametric [L3; Parisi, 1979, 1980; Mézard et al., 1987]. For any three pure states α, β, γ, the overlaps $q_{\alpha\beta} = \frac{1}{N}\sum_i \langle s_i \rangle_\alpha \langle s_i \rangle_\beta$ satisfy:

$$q_{\alpha\beta} \geq \min(q_{\alpha\gamma}, q_{\beta\gamma})$$

This is equivalent to the ultrametric inequality for a distance defined as $d_{\alpha\beta} = 1 - q_{\alpha\beta}$. Parisi's result was not a search for tree structure—it emerged from the mathematics of replica symmetry breaking and was subsequently recognized as ultrametricity.

Triadic rigidity test: For any three pure states, the two largest overlaps are equal. This is a testable prediction of the RSB solution and has been confirmed in numerical simulations.

Resolution-dependence: Temperature acts as an inverse resolution parameter. At high temperature (coarse resolution), only a few states are distinguishable. As temperature decreases (finer resolution), the state space differentiates into an ultrametric tree of pure states. The transition from paramagnetic to spin glass phase is the onset of tree structure—the first cut in the dendrogram.

Evidence strength: L3 (published, replicated, widely accepted). This is the strongest evidence for ultrametric organization outside the QWAV program. Parisi was awarded the 2021 Nobel Prize in Physics for this work.

Noun/verb in spin glasses: The ultrametric topology of the state space is the ontic invariant. The specific numerical values of $q_{\alpha\beta}$ are epistemic—they depend on the specific realization of disorder in the Hamiltonian. What persists across realizations is the tree structure, not the overlap values.

3.3 Protein Folding [L4]

Structural analogy with strong theoretical support.

The energy landscape theory of protein folding posits that the energy surface of a folding protein is organized as a hierarchical funnel [L4; Bryngelson & Wolynes, 1987; Frauenfelder et al., 1991; Onuchic et al., 1997]. Conformations that are structurally similar lie in the same basin; basins are nested within larger basins at coarser levels of structural similarity. The native state sits at the funnel's minimum.

Triadic rigidity test [L4—not yet empirically verified]: For any three conformations, define distance by the structural similarity at which their folding pathways diverge (the height of their LCA in the folding funnel). If the funnel is genuinely ultrametric, the two largest pairwise distances should be equal.

Resolution-dependence: The depth of the folding funnel is a resolution parameter. At coarse resolution (early folding), only broad structural classes are distinguished. At fine resolution (late folding), individual conformations are discriminated. The folding process is a descent through the tree from root (unfolded ensemble) to leaf (native state).

Evidence strength: L4. The structural analogy is well-motivated and consistent with energy landscape theory, but direct empirical tests of triadic rigidity in folding pathways have not been performed. The analogy is supported by the success of hierarchical clustering methods in protein structure classification (CATH, SCOP databases).

Noun/verb in protein folding: The funnel topology—which basins contain which sub-basins—is the ontic invariant. Specific folding energies and rates are sequence-dependent (epistemic). The tree structure describes the *topology of possible pathways*; the specific trajectory a given molecule takes is a realization within that topology.

3.4 Cosmology [L5]

Suggestive structural patterns.

The Tree Distance Cophenetic paper proposed a cosmic tree interpretation [L5; Quni-Gudzinas, 2026c, §3.3]: the history of the universe from the Big Bang to the present can be viewed as a process of successive differentiation—a tree whose root is the pre-Big Bang state and whose branching events correspond to symmetry breaking, force separation, and structure formation: inflation $\rightarrow$ electroweak $\rightarrow$ quark-gluon plasma $\rightarrow$ hadronization $\rightarrow$ nucleosynthesis $\rightarrow$ recombination $\rightarrow$ structure formation $\rightarrow$ present complexity.

Triadic rigidity test [L5—speculative]: If cosmic history is tree-structured in the cophenetic sense, then for any three cosmic structures (galaxies, clusters, etc.), the two largest "formation divergence times" should be equal. This is testable in principle through correlation function analysis, but no such test has been performed.

Resolution-dependence: Cosmic time functions as a resolution parameter. The root (pre-BB) is the coarsest description—no distinctions exist. Each symmetry-breaking event creates a new branching level, differentiating the universe at finer resolution. The present moment is the leaf set—maximum differentiation.

Evidence strength: L5 (suggestive). The structural analogy is coherent and not contradicted by known cosmology, but it has not been subjected to the triadic rigidity test. This is the weakest domain in the synthesis, included for completeness and as a direction for future investigation, not as an established result.

Noun/verb in cosmology: The branching order—which structures differentiate from which, in what sequence—would be the ontic invariant if the cosmic tree interpretation holds. Specific times, energy scales, and physical constants are epistemic.

3.5 Cognition and Linguistics [L5]

Suggestive structural patterns.

Taxonomic hierarchies in both natural and constructed classification systems exhibit tree structure: biological taxonomy (Linnaean), language families (Indo-European tree), semantic networks (WordNet hierarchies), and conceptual ontologies. These are explicitly

constructed as trees, suggesting that hierarchical organization is a natural mode of human cognition [L5; Quni-Gudzinas, 2026c, §4.1].

Triadic rigidity test [L5—speculative]: For any three concepts in a semantic hierarchy, the two largest semantic distances (measured by the depth of their lowest common hypernym) should be equal. This is testable with existing lexical databases (WordNet) but has not been systematically evaluated.

Resolution-dependence: Category granularity functions as resolution. At coarse resolution (high in the hierarchy), only broad categories are distinguished (animal vs. plant). At fine resolution (near the leaves), specific concepts are discriminated (golden retriever vs. Labrador). The hierarchy is navigated at different resolutions depending on task context.

Evidence strength: L5 (suggestive). The existence of tree-structured taxonomies is not in dispute—it is a design feature of how we organize knowledge. What is unproven is whether these taxonomies reflect an underlying ultrametric structure in semantic space (as opposed to being a convenient organizing convention). The triadic rigidity test would distinguish these possibilities.

Noun/verb in cognition: The tree topology of a semantic hierarchy is the ontic invariant if ultrametric structure holds. Specific semantic similarity scores and distance values from particular embedding models are epistemic.

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4. Objections

We address seven objections from the Tree Distance Cophenetic framework [Quni-Gudzinas, 2026c, §6], adapted to the cross-domain context.

O1: “This is just hierarchical clustering—a data analysis technique.”

Response: Hierarchical clustering is an algorithm that *produces* a dendrogram from any distance matrix. The present claim is the reverse: that the underlying structure of certain physical systems *is* a tree with cophenetic distance, and that hierarchical clustering would recover it because it is there, not because the algorithm imposes it. The distinction is between “we clustered the data and got a tree” and “the system’s intrinsic geometry is a tree—and here is a falsifiable test.” Triadic rigidity provides the test: if the system is genuinely tree-structured, triadic rigidity holds independent of clustering algorithm. If it is not, no clustering algorithm can make it so.

O2: “What about non-hierarchical phenomena? Not everything is a tree.”

Response: Correct—and the framework does not claim otherwise. The claim is bounded: *where* hierarchical organization exists, it has a specific mathematical signature. Triadic rigidity is both the signature and the boundary. If triadic rigidity fails, the framework does not apply to that domain, at that resolution, for those items. This is a feature, not a bug—it

prevents the framework from being unfalsifiable. The honest response to "not everything is a tree" is: "Agreed. Here is how we tell which things are."

O3: "Why privilege binary distinctions? Real hierarchies branch n-way."

Response: The framework does not privilege binary branching. The proof of the ultrametric inequality and triadic rigidity holds for trees with arbitrary branching factors (binary, ternary, n-ary). The ternary tree ($p = 3$) used in the quantum error correction architecture is chosen for symmetry reasons [L2; Quni-Gudzinas, 2026b], not because trees must be ternary. The framework accommodates any branching structure—what matters is the nesting property, not the branching factor.

O4: "The cosmological interpretation is unfalsifiable—we cannot re-run the universe."

Response: The cosmic tree interpretation is L5—we are explicit that it is suggestive, not proven. However, it is not unfalsifiable in principle. Triadic rigidity makes a specific prediction about the correlation structure of cosmic observables. If CMB or large-scale structure data were to systematically violate triadic rigidity, the cosmic tree interpretation would be falsified. The limitation is data, not principle.

O5: "Parisi's ultrametricity is a property of the *mean-field* solution—it does not hold in finite dimensions."

Response: This is an active area of research. The mean-field SK model is exactly solvable and yields ultrametricity. Finite-dimensional spin glasses (Edwards–Anderson model) are not exactly solvable, and the question of whether ultrametricity survives in finite dimensions is unresolved. We acknowledge this: the spin glass evidence is L3 (published result for the mean-field case), not L1 (theorem for all cases). The framework does not require ultrametricity to hold universally in all spin glasses—it requires that where it holds, it has the cophenetic signature.

O6: "Cognition and linguistics are human constructs—finding trees there just means we like to organize things hierarchically."

Response: This is a substantive objection that the framework does not dismiss. It is possible that taxonomic hierarchies reflect cognitive convenience rather than an underlying ultrametric structure in semantic space. The triadic rigidity test would distinguish: if semantic distances are genuinely ultrametric, triadic rigidity should hold for concept triples independent of the specific taxonomy used. If taxonomic trees are merely convenient fictions, triadic rigidity should fail for carefully constructed test sets. This is an empirical question, and we flag it as L5 precisely because the test has not been done.

O7: "The framework is too abstract—it maps onto everything because it says nothing specific about anything."

Response: This objection would be valid if the framework were unfalsifiable. It is not. Triadic rigidity is a specific, testable mathematical signature. For quantum error correction, it has been verified computationally [L2]. For spin glasses, it is a known consequence of

the RSB solution [L3]. For the remaining domains, it makes predictions that have not yet been tested. A framework that maps onto everything without saying anything is unfalsifiable. A framework that says "if the domain is tree-structured, this specific inequality must hold" is falsifiable—it says something concrete, and that thing can be wrong.

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5. Falsifiability and Next Steps

5.1 Triadic Rigidity as Universal Test

The central claim of this synthesis is falsifiable. For each domain, we specify:

Domain	Falsification Condition	Status
Quantum error correction	Triadic rigidity fails for error syndrome distances at any depth d	Tested: holds at all tested depths [L2]
Spin glasses (mean-field)	Overlap structure violates $q_{\alpha\beta} \geq \min(q_{\alpha\gamma}, q_{\beta\gamma})$	Tested: holds in SK model [L3]
Spin glasses (finite-dim)	Triadic rigidity fails for state overlaps in EA model simulations	Open question
Protein folding	Triadic rigidity fails for folding pathway divergences	Not yet tested [L4]
Cosmology	Correlation structure of cosmic observables violates triadic rigidity	Not yet tested [L5]
Cognition/linguistics	Semantic distances for concept triples violate triadic rigidity	Not yet tested [L5]

If triadic rigidity fails in any domain, the ultrametric thesis is falsified *for that domain*—not for the framework as a whole. This is the bounded consilience principle in action: the framework is a family of domain-specific hypotheses, not a single monolithic claim.

5.2 Open Questions

Beyond domain-specific falsification, several open questions from the broader QWAV program bear on this synthesis [Quni-Gudzinas, 2026e]:

1. **p-adic Solovay–Kitaev problem:** Can a small set of tree automorphisms generate a dense subset of the unitary group, providing a universal gate set native to the tree geometry?
2. **Physical realizability:** Can the tree's energy landscape be physically built, or does it require structures (perfect tensors, AME states) that may not exist for arbitrary parameters?

3. **Perfect tensor existence:** Do absolutely maximally entangled (AME) states exist for every number of parties and local dimension? The tree architecture requires them.
4. **Tree automorphism gate set:** Proof that tree automorphisms form a universal gate set for quantum computing on the Bruhat–Tits tree.

These questions connect the computational validation (Section 3.1) to the broader physical program. The synthesis does not depend on their resolution, but their resolution would either strengthen or constrain the framework.

5.3 Contribution and Limitations

What this synthesis contributes: - A unified mathematical framework (cophenetic distance + ultrametric inequality) for understanding hierarchical organization across domains - A falsifiable signature (triadic rigidity) that separates genuine tree structure from metaphorical tree-invocation - An honesty taxonomy (L1–L5) that prevents overclaiming while preserving the connective tissue between domains - A resolution-dependence lens that reconciles Archimedean and ultrametric descriptions as descriptions at different scales - A noun/verb distinction that separates ontic invariants from epistemic conventions

What this synthesis does not contribute: - New computational or experimental results in any single domain - A proof that ultrametric geometry is the *only* geometry of hierarchical organization - A demonstration that all five domains are ultrametric at L1–L2 confidence - A claim that non-hierarchical phenomena do not exist or are reducible to trees

The synthesis is a map of connective tissue. It shows that the same mathematical structure recurs, at varying levels of evidential support, across domains that are rarely discussed together. Whether this recurrence reflects a deep unity or a series of coincidences is an empirical question—and the framework provides the tools to answer it.

Appendix A: Citation Inventory

DOI	Reference	Domain	Level
`10.5281/zenodo.20213043`	Quni-Gudzinas, *Tree Distance Cophenetic* (2026c)	Mathematical foundation	L1
`10.5281/zenodo.20208437`	Quni-Gudzinas, *Symmetric Extension* (2026b)	Quantum error correction	L2
`10.5281/zenodo.20134944`	Quni-Gudzinas, *Tier 0 Validation* (2026a)	Quantum error correction	L2
`10.5281/zenodo.20154557`	Quni-Gudzinas, *Ultrametric Quantum Computing Foundations* (2026)	Quantum error correction	L1–L2

DOI	Reference	Domain	Level
`10.5281/zenodo.20061155`	Quni-Gudzinas, *How Geometry Creates Memory* (2026d)	Threshold principle	L1
`10.5281/zenodo.20089407`	Quni-Gudzinas, *A Different Geometry for Computing* (2026e)	Open questions	L1–L2
`10.5281/zenodo.20095901`	Quni-Gudzinas, *Adelic Constraints Phase 1* (2026)	Number theory bridge	L2
—	Parisi, *Phys. Rev. Lett.* 43, 1754 (1979)	Spin glasses	L3
—	Mézard, Parisi & Virasoro, *Spin Glass Theory and Beyond* (1987)	Spin glasses	L3
—	Bryngelson & Wolynes, *PNAS* 84, 7524 (1987)	Protein folding	L4
—	Frauenfelder et al., *Science* 254, 1598 (1991)	Protein folding	L4

Appendix B: Consilience Level Taxonomy

Level	Definition	Burden of Proof	Example in This Paper
L1	Mathematical theorem	Derivation from axioms	Ultrametric inequality $\Longrightarrow$ triadic rigidity
L2	Computational validation	Reproducible numerical experiment	Zero logical errors at $d = 7$, $3^7 = 2{,}187$ leaves
L3	Published result (other researchers)	Peer-reviewed, replicated	Parisi RSB: spin glass state space is ultrametric
L4	Structural analogy	Clean mapping, no contradiction with known results	Protein folding funnel as ultrametric tree
L5	Suggestive pattern	Coherent with framework, not yet subjected to rigorous test	Cosmic timeline as successive differentiation tree

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Draft v0.3 — FHT Integration (Revised). Written 2026-05-18. All claims labeled [L1]–[L5]. Triadic rigidity verified computationally in QEC domain; pending empirical test in protein folding, cosmology, and cognition domains.